Tutorial on Post-APR Procedures

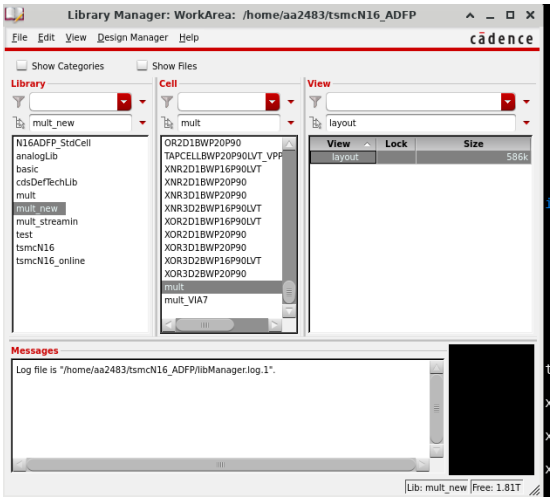
After completing APR, the post-APR Verilog netlist is saved and the GDS file is exported. Next, we stream the GDS file into Cadence Virtuoso.

1) **First, we need to import the *.gds file as a layout view in Cadence Virtuoso.**
This is done as follows:

- First, close any virtuoso session if you have it open
- Open cds.lib file from your ~/tsmcN16_ADFP directory.
- Add the following line to your cds.lib file, if not present:

DEFINE tsmcN16_online /classes/ece5746/tsmcN16/Executable_Package/Collaterals/Tech/iPDK/N16ADFP_iPDK/Techfile/online/1P11M_2Xa1Xd_h_3Xe_vhv_2Y2R/techfile_lib_laker/tsmcN16/

- Copy the file '**streamIn.sh**' from the path of **/classes/ece5746/projects/TutorialAPR/streamIn.sh** to your local folder of **~/tsmcN16_ADFP**
- In your '**streamIn.sh**' file, do the following:
  - Change the path of -strmFile option to point to your gds generated from APR.
  - Change the -topCell option name to your design module name
  - Change the -library option value to the new library that you want to create
- Source the '**streamIn.sh**' in your **~/tsmcN16_ADFP** directory using source command:
  - source streamIn.sh
- Open Virtuoso using the following command:
  - virtuoso &
- When you run Virtuoso, you should see the new library has been added in your library list and it will have the layout view of your design. Open and check the layout view of your design.



Streamed in library named mult_new. Open layout by selecting mult cell and layout view.

2) **SDF back-annotated RTL simulation**

An SDF file is referred to as Standard Delay Format file, which contains real timing information about the standard cells and interconnects in your layout. We perform SDF back-annotated simulation to apply realistic delays in our RTL simulation.

Considering the real delays of the cells and interconnects, the clock edge where a data is triggered might appear slightly different (early or late) than what we assume with ideal simulations. This might break you design with unforeseen timing violations which are uncertain. Hence this check is the final check to verify the proper functionality of your chip with real simulation.

- Following the Tutorial on RTL simulation create a new project with testbench for your design
- Include the <design>.<clock_period>.apr.v Verilog netlist generated from APR in this newly created project.
- Copy the following standard cell RTL file into your simulation run directory, add it in your simulation project
    - /classes/ece5746/tsmcN16/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/VERILOG/N16ADFP_StdCell.v
- Add the following code block in your test bench

    initial begin

        $sdf_annotate("<path to sdf file>",<testbench module name>.<instance name>,,"sdf_log.log","maximum");

    end
- Compile and simulate and check waveforms.

```
initial begin
        $sdf_annotate("/home/aa2483/tsmcN16_ADFP/lab2/apr/apr_clean/output/mult.2.apr.sdf",mult_tb.dut,,"sdf_log.log","maximum");
end
```

An example inclusion of $sdf_annotate in testbench.